SANGRAMJEET GHOSH

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neuroviai.dev | jeetintech.github.io | Verified Certificates Portfolio
Hyderabad, Telangana, India | Right to work: Requires UK Skilled Worker visa sponsorship

Professional Summary

AI & Full-Stack Software Engineer (B.Tech CSE AI/ML, 2026) who built and runs Neuroviai, a live SaaS with paying users, and published a VS Code extension with 904 installs. Seeking graduate software/AI engineering roles in the UK; open to relocation.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering *Graduated May 2026*
Specialization: **Artificial Intelligence and Machine Learning** **GPA: 8.0 / 10.0**
Malla Reddy University, Hyderabad, Telangana, India

- **Relevant Modules:** Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Database Management Systems, Cloud Computing, Distributed Systems, Probability & Statistics.
- **Undergraduate Dissertation / Thesis:** *Design and Implementation of Cost-Constrained AI Automation Systems* (Feb 2026) — Full-stack SaaS platform (Neuroviai) spanning 15,000+ lines of Python and 12,000+ lines of TypeScript; implemented regulatory-compliant data stewardship aligning with India's DPDPA 2023 and EU AI Act principles.

Technical Competencies

Programming Languages	Python (3.10–3.13), TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3/PostCSS, Rust (Basics), Bash
Agentic AI & LLMs	LangGraph (`StateGraph`), LangChain, Model Context Protocol (MCP SDK), Gemini Live API, Groq (Llama-3), OpenAI API, Ollama, Prompt Engineering, Multi-Agent ReAct Loops
Vector DBs & Memory	PostgreSQL (`pgvector` Cosine Similarity), ChromaDB, FAISS, SQLite Finite State Machines, Knowledge Graphs
Computer Vision & Audio	OpenCV, MediaPipe, Librosa (Acoustic Prosody), Real-ESRGAN, Whisper/faster-whisper, Kokoro TTS, Edge-TTS, FFmpeg CLI
Backend & Microservices	FastAPI, AsyncIO, RESTful APIs, WebSockets, Pydantic v2, Celery, Redis Task Queues, Netlify Functions, JWT, OAuth 2.0, pytest
Frontend Development	React 18/19, Next.js 14/16 (App Router, Turbopack), Tailwind CSS (v3/v4), Zustand, Framer Motion, Webview IPC
Databases & Storage	PostgreSQL, Supabase (Row-Level Security, Realtime, Triggers), SQLite, Redis, AWS DynamoDB, Amazon S3
Cloud & DevOps	AWS (EC2, S3, Lambda, IAM), Docker, Docker Compose, Git, GitHub Actions CI/CD, Nginx, Render, Netlify, Vercel
Developer Tooling	VS Code Extension API, Playwright, Headless Chrome (CDP), PySide6 (Qt6), Monaco Editor, Postman

Professional Experience

Founding Engineer *Sep 2026 – Present*
Molars Nutrition Brand Platform Hyderabad, India

- Built the brand platform end-to-end using React 19, Next.js, TypeScript, and Tailwind CSS.
- Engineered custom interactive molecular canvas components (`molecule.tsx`) with physics-based cursor repulsion, sustaining a steady 60 FPS across mobile and desktop.
- Implemented serverless Netlify Functions (`waitlist.mts`) with transactional email verification, client-side input sanitization, and automated Supabase database synchronization.
- Optimized page delivery to achieve 95+ Google Lighthouse scores across Performance, Accessibility, Best Practices, and SEO.

Freelance Software Engineer *Aug 2025 – Present*
Independent Remote

- Designed, engineered, and delivered full-stack web platforms and bespoke AI-powered MVPs for 3+ international startup founders from initial concept to cloud deployment.
- Implemented subscription token billing via Razorpay webhooks, role-based access control (JWT/OAuth 2.0), and multi-platform publishing integrations.
- Orchestrated multi-tier containerized deployments across AWS EC2, Render, Vercel, and Netlify with custom domain DNS, SSL certificates, and Cloudflare CDN caching.
- Formulated LLM cost-containment architectures utilizing multi-provider failover routing, response caching, and prompt token budgeting, reducing client inference expenditures by up to 60%.

Data Science & Backend Intern *Mar 2025 – Dec 2025*
Inikola Technologies Hyderabad, India

- Engineered high-throughput backend services and NLP scoring algorithms for CraftingBrain (craftingbrain.com), a live commercial content-technology platform, using Python and FastAPI.
- Built automated data validation and ingestion pipelines with Pydantic and PostgreSQL, sanitizing high-volume unstructured text feeds.
- Administered AWS cloud infrastructure: provisioned EC2 instances, structured S3 asset access policies, configured IAM roles, and implemented Nginx reverse proxies with SSL termination.
- Containerized microservices using multi-stage Docker builds (cutting footprint by 45%) and implemented GitHub Actions CI/CD for linting, testing, and zero-downtime releases.

Key Technical Projects & Systems Architecture

Neuroviai — AI-Powered Multi-Service Content Automation SaaS *Mar 2025 – Present*
Next.js 14, TypeScript, FastAPI, PostgreSQL, Supabase, Celery, Redis, Razorpay, Docker neuroviai.dev

- Architected a live commercial content automation SaaS platform currently generating **1.2K+ organic monthly visitors** with active paying customers, via 6 decoupled FastAPI microservices managing identity, billing, media generation, and scheduling.
- Engineered a 4-tier circuit-breaking LLM gateway (Gemini Pro → Groq Llama-3 → HuggingFace → Local Ollama) achieving 99.7% uptime, buffering burst traffic within strict rate limits (12k tokens/min).
- Designed an atomic token billing economy (20× redenomination) secured via Razorpay HMAC-SHA256 signature verification and PostgreSQL stored procedures, preventing double-spending anomalies.
- Implemented a custom client-side `ServerWarmer` pattern in Next.js 14 App Router, cutting cold-start latency from 5.2s to sub-150ms, and isolated heavy browser automation into Celery/Redis background queues.

Sebastian — Autonomous AI Operating System & Private Digital Butler *Jan 2026 – Present*
Python, Google Gemini Live API, Kokoro Neural TTS, PostgreSQL, pgvector, MCP SDK, PyAudio github.com/JeetInTech

- Engineered a 25-module autonomous personal operating system featuring full-duplex bidirectional streaming audio via Gemini Live (∼400ms turnaround) with barge-in interruption detection.
- Orchestrated 92 registered tools across Green (read-only) and Yellow (action-required) permission gates, managing dynamic `npx` child-process Model Context Protocol (MCP) servers (GitHub, Playwright, Filesystem).
- Engineered turn-by-turn semantic memory retrieval powered by **PostgreSQL + `pgvector` cosine similarity (sub-10ms)** and persistent Knowledge Graph entity correlation, plus a Discord voice bridge resampling audio between Discord (48kHz stereo) and the Gemini Live pipeline (16kHz mono).

Mr. Memecredible — Published VS Code Extension *Published Nov 2025*
TypeScript, VS Code Extensibility API, Node.js, Webview Panels, JSON Diagnostics 904 Active Installs

- Developed and published a real-time developer productivity extension on the official Microsoft Visual Studio Marketplace, amassing **904 active developer installs**.
- Built an asynchronous, non-blocking diagnostic listener subscribing to `vscode.languages.onDidChangeDiagnostics` within a strict <16ms budget, driving bi-directional postMessage IPC to isolated Webview panels rendering a 9-stage uncanny meme meter.
- Maintained an ultra-compact bundle size under 250KB with zero external runtime dependencies.

Genesis — Master Fleet Operations Hub & Multi-Agent Process Supervisor *Aug 2026 – Present*
Python, PySide6 (Qt6), SQLite, psutil, NumPy, Windows Process API, Regex Incident Engine github.com/JeetInTech

- Engineered a native desktop control plane supervising 10 autonomous background agents, verifying system health through real evidence (process trees, lock files, log recency) rather than self-reported heartbeats.
- Built an evidence-based entity topology engine linking independent agents when shared entities traverse data stores, and a deterministic regex incident engine intercepting fatal crash logs to trigger automatic recovery.
- Vectorized a 60,000-particle animated visualization in pure NumPy at 60 FPS without blocking the GUI thread.

AIExamTool — Automated Examination & Evaluation Platform (Final Year Project) *Jan – Apr 2026*
Python, AWS Lambda, DynamoDB, FAISS, OpenAI API, Flask, ReportLab github.com/JeetInTech

- Built a serverless academic evaluation platform with role-scoped portals (student, faculty, HOD) on AWS Lambda and DynamoDB.
- Implemented FAISS-backed semantic search over course syllabi for grounded question generation and automated semantic answer evaluation.

Additional Projects

- **AI Local Video Editor**: 100% offline editor using local GPU speech transcription to trim dead air and auto-insert memes via FFmpeg.
- **Genesis: Job Application Agent**: Playwright agent that parses listings, tailors LaTeX resumes via Groq LLM, and tracks applications across 5 job portals.
- **Virtual Therapist**: LangGraph mental-wellness agent combining facial/voice analysis with rule-based safety guardrails that run before any LLM call.
- **Genesis: Video Pipeline**: Autonomous YouTube Shorts pipeline for script generation, TTS voiceover, and FFmpeg compositing via the YouTube Data API v3.
- **OpenInstaFlow**: MCP server (TypeScript) exposing the Instagram Graph API to Claude Desktop and Cursor.
- **Realtime Multiverse**: Real-time community platform in React 19, Next.js 16 (Turbopack), and Supabase Realtime WebSockets.
- **Local Cortex**: Native desktop AI IDE (Tauri v2, Rust) with local vector RAG over Ollama.
- **PyPI Package (`jeet`)**: Open-source developer utility published via Twine and Setuptools.

Achievements & Leadership

- **Winner — IDEATHON 2.0** (University-Level Ideation Competition): automated AI healthcare solution pitched to industry judges on architecture, feasibility, and market viability.
- **Participant — METALLOOP 2K23** (36-hour national hackathon): built and pitched a working prototype.
- **Open-Source Contributor**: Actively maintain 30+ public repositories on GitHub (github.com/JeetInTech) across AI systems, developer tooling, and web frameworks.

Certifications *(verified portfolio)*: AWS Cloud Practitioner Essentials (course) | GCP Cloud Computing Fundamentals | DeepLearning.AI ML Specialization & AI Foundations | Azure AI Fundamentals (AI-900, learning path) | NPTEL Programming & DSA • **Languages**: English (Professional), Hindi (Native), Bengali (Native)